4a_population_pca

Auto-generated from `2_pipeline/4a_population_pca/out/` on 2026-09-14 16:52. Re-run `make_progress_md.py` after the script's output changes -- don't hand-edit this file.

Tables

Pca real summary

session_id	window	roi_name	labeled_grouped	n_cells	n_trials	v
LPE10884_2024_01_12	pre_stim	AL	unl	320	209	0
LPE10884_2024_01_12	stim	AL	unl	320	209	0
LPE10884_2024_01_12	reward	AL	unl	320	209	0
LPE10884_2024_01_12	pre_stim	RSP	unl	63	209	0
LPE10884_2024_01_12	stim	RSP	unl	63	209	0
LPE10884_2024_01_12	reward	RSP	unl	63	209	0
LPE10884_2024_01_17	pre_stim	AL	unl	417	258	0
LPE10884_2024_01_17	stim	AL	unl	417	258	0
LPE10884_2024_01_17	reward	AL	unl	417	258	0
LPE10884_2024_01_17	pre_stim	RSP	unl	159	258	0
LPE10884_2024_01_17	stim	RSP	unl	159	258	0
LPE10884_2024_01_17	reward	RSP	unl	159	258	0
LPE11622_2024_02_20	pre_stim	PM	lab	220	396	0
LPE11622_2024_02_20	stim	PM	lab	220	396	0
LPE11622_2024_02_20	reward	PM	lab	220	396	0
LPE11622_2024_02_20	pre_stim	PM	unl	740	396	0
LPE11622_2024_02_20	stim	PM	unl	740	396	0
LPE11622_2024_02_20	reward	PM	unl	740	396	0
LPE11622_2024_02_20	pre_stim	V1	lab	99	396	0
LPE11622_2024_02_20	stim	V1	lab	99	396	0
LPE11622_2024_02_20	reward	V1	lab	99	396	0
LPE11622_2024_02_20	pre_stim	V1	unl	495	396	0
LPE11622_2024_02_20	stim	V1	unl	495	396	0
LPE11622_2024_02_20	reward	V1	unl	495	396	0
LPE11622_2024_02_22	pre_stim	AL	unl	105	438	0

session_id	window	roi_name	labeled_grouped	n_cells	n_trials	
LPE11622_2024_02_22	stim	AL	unl	105	438	
LPE11622_2024_02_22	reward	AL	unl	105	438	
LPE11622_2024_02_22	pre_stim	PM	lab	120	438	
LPE11622_2024_02_22	stim	PM	lab	120	438	
LPE11622_2024_02_22	reward	PM	lab	120	438	

(showing first 30 of 339 rows -- see the linked CSV for the rest)

pca_real_summary.csv

Pca shuffle results

session_id	window	roi_name	labeled_grouped	n_cells	condition
LPE10884_2024_01_12	pre_stim	AL	unl	320	real
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	global
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim

session_id	window	roi_name	labeled_grouped	n_cells	condition
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim
LPE10884_2024_01_12	pre_stim	AL	unl	320	stim

(showing first 30 of 27459 rows -- see the linked CSV for the rest)

pca_shuffle_results.csv

Trajectory scores

session_id	roi_name	labeled_grouped	s_rel	PC1	PC2	PC3
LPE10884_2024_01_12	AL	unl	-75	3.342	-6.092	2.57
LPE10884_2024_01_12	AL	unl	-65	-4.243	-1.669	3.22
LPE10884_2024_01_12	AL	unl	-55	-7.125	-3.919	-5.1
LPE10884_2024_01_12	AL	unl	-45	-8.155	-0.1767	-2.5
LPE10884_2024_01_12	AL	unl	-35	-7.892	6.167	-5.9
LPE10884_2024_01_12	AL	unl	-25	-3.976	7.761	-4.9
LPE10884_2024_01_12	AL	unl	-15	4.388	9.067	-6.6
LPE10884_2024_01_12	AL	unl	-5	11	8.076	0.44
LPE10884_2024_01_12	AL	unl	5	13.68	6.006	5.16
LPE10884_2024_01_12	AL	unl	15	16.83	-0.9472	4.63
LPE10884_2024_01_12	AL	unl	25	8.279	-19.89	-9.6
LPE10884_2024_01_12	AL	unl	35	-7.79	-5.587	15.4
LPE10884_2024_01_12	AL	unl	45	-10.49	0.3235	6.23
LPE10884_2024_01_12	AL	unl	55	-7.851	0.8771	-2.8
LPE10884_2024_01_12	RSP	unl	-75	1.371	-2.365	1.07
LPE10884_2024_01_12	RSP	unl	-65	-1.9	-3.41	1.17
LPE10884_2024_01_12	RSP	unl	-55	-4.402	-0.416	-0.2
LPE10884_2024_01_12	RSP	unl	-45	-2.959	0.3309	0.08
LPE10884_2024_01_12	RSP	unl	-35	-4.845	6.576	-1.6

session_id	roi_name	labeled_grouped	s_rel	PC1	PC2	PC3
LPE10884_2024_01_12	RSP	unl	-25	-2.185	4.467	-2.2
LPE10884_2024_01_12	RSP	unl	-15	-0.4707	4.546	-1.6
LPE10884_2024_01_12	RSP	unl	-5	7.75	3.974	5.15
LPE10884_2024_01_12	RSP	unl	5	8.672	1.633	1.13
LPE10884_2024_01_12	RSP	unl	15	5.94	-1.51	-5.5
LPE10884_2024_01_12	RSP	unl	25	2.929	-6.388	-3.5
LPE10884_2024_01_12	RSP	unl	35	-2.849	-5.626	2.75
LPE10884_2024_01_12	RSP	unl	45	-3.415	-1.072	1.90
LPE10884_2024_01_12	RSP	unl	55	-3.638	-0.7409	1.58
LPE10884_2024_01_17	AL	unl	-75	5.265	2.722	4.20
LPE10884_2024_01_17	AL	unl	-65	0.7212	4.82	1.97

(showing first 30 of 1582 rows -- see the linked CSV for the rest)

trajectory_scores.csv

Trajectory variance

session_id	roi_name	labeled_grouped	PC	var_ratio	n_cells	n_
LPE10884_2024_01_12	AL	unl	1	0.2536	320	14
LPE10884_2024_01_12	AL	unl	2	0.1709	320	14
LPE10884_2024_01_12	AL	unl	3	0.1299	320	14
LPE10884_2024_01_12	AL	unl	4	0.07925	320	14
LPE10884_2024_01_12	AL	unl	5	0.05566	320	14
LPE10884_2024_01_12	RSP	unl	1	0.3109	63	14
LPE10884_2024_01_12	RSP	unl	2	0.2225	63	14
LPE10884_2024_01_12	RSP	unl	3	0.111	63	14
LPE10884_2024_01_12	RSP	unl	4	0.08965	63	14
LPE10884_2024_01_12	RSP	unl	5	0.05464	63	14
LPE10884_2024_01_17	AL	unl	1	0.2949	417	14
LPE10884_2024_01_17	AL	unl	2	0.221	417	14
LPE10884_2024_01_17	AL	unl	3	0.1071	417	14
LPE10884_2024_01_17	AL	unl	4	0.06443	417	14
LPE10884_2024_01_17	AL	unl	5	0.04943	417	14
LPE10884_2024_01_17	RSP	unl	1	0.3571	159	14

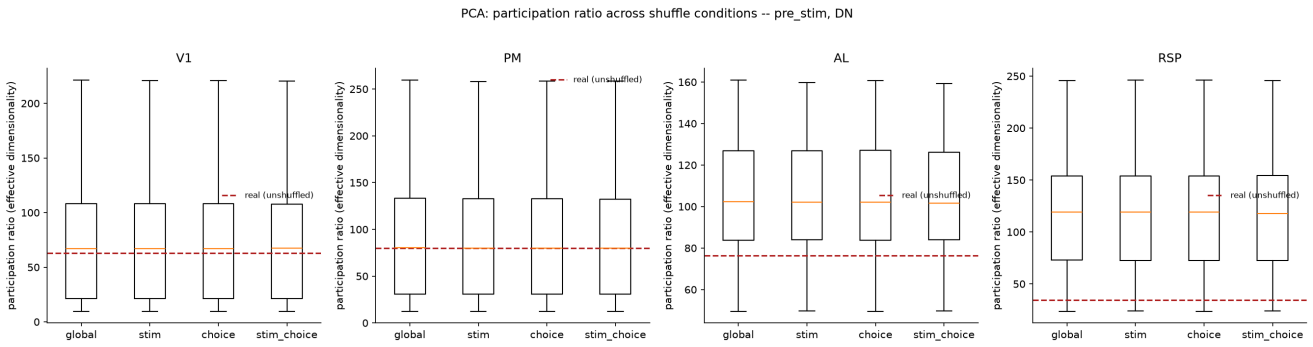
session_id	roi_name	labeled_grouped	PC	var_ratio	n_cells	n_
LPE10884_2024_01_17	RSP	unl	2	0.2096	159	14
LPE10884_2024_01_17	RSP	unl	3	0.09608	159	14
LPE10884_2024_01_17	RSP	unl	4	0.07216	159	14
LPE10884_2024_01_17	RSP	unl	5	0.04534	159	14
LPE11622_2024_02_20	PM	lab	1	0.3357	220	14
LPE11622_2024_02_20	PM	lab	2	0.2234	220	14
LPE11622_2024_02_20	PM	lab	3	0.09565	220	14
LPE11622_2024_02_20	PM	lab	4	0.06557	220	14
LPE11622_2024_02_20	PM	lab	5	0.04499	220	14
LPE11622_2024_02_20	PM	unl	1	0.3203	740	14
LPE11622_2024_02_20	PM	unl	2	0.2141	740	14
LPE11622_2024_02_20	PM	unl	3	0.09478	740	14
LPE11622_2024_02_20	PM	unl	4	0.06699	740	14
LPE11622_2024_02_20	PM	unl	5	0.04505	740	14

(showing first 30 of 565 rows -- see the linked CSV for the rest)

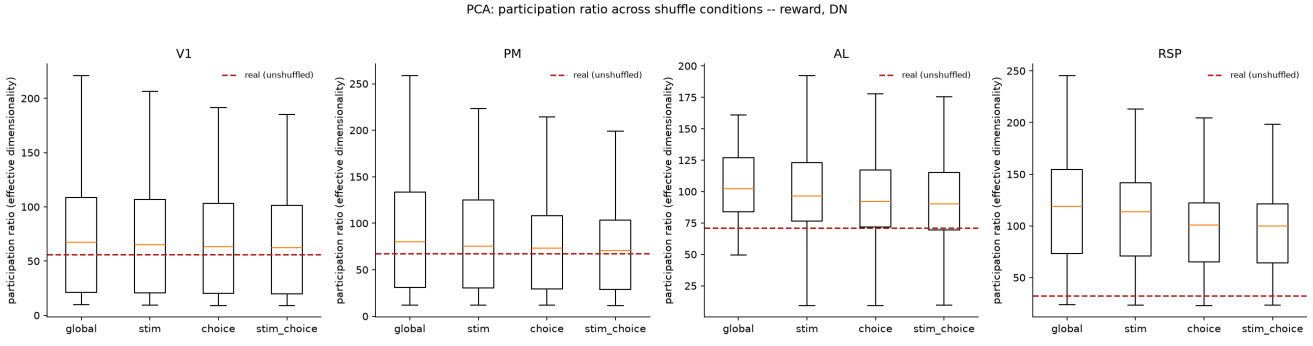
trajectory_variance.csv

Figures

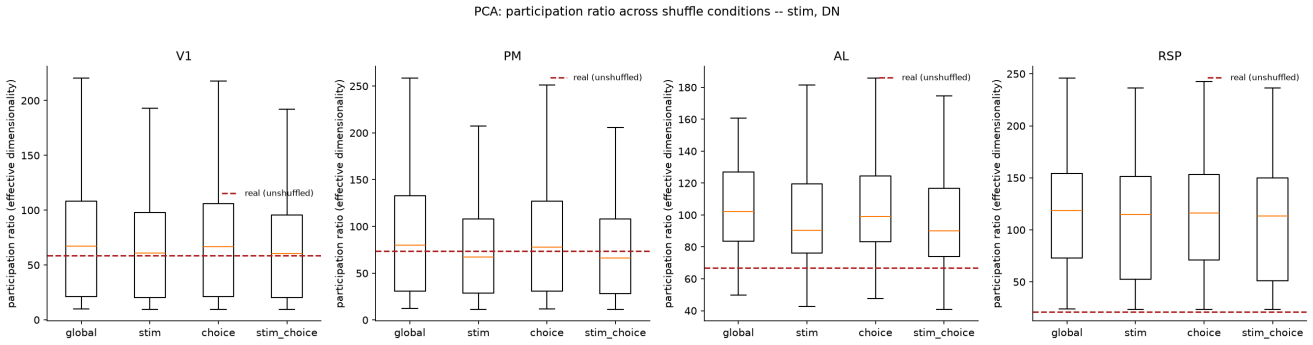
Shuffle comparison pre stim



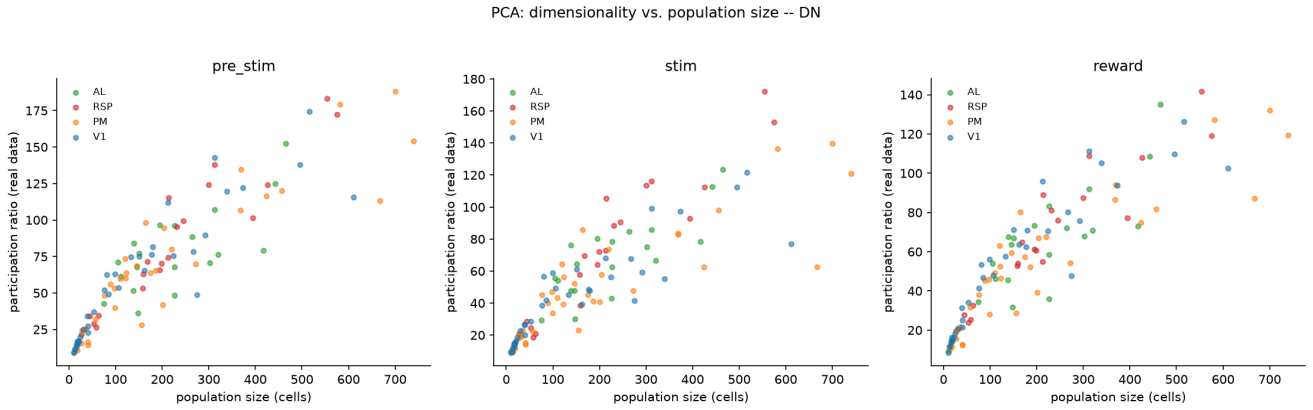
Shuffle comparison reward



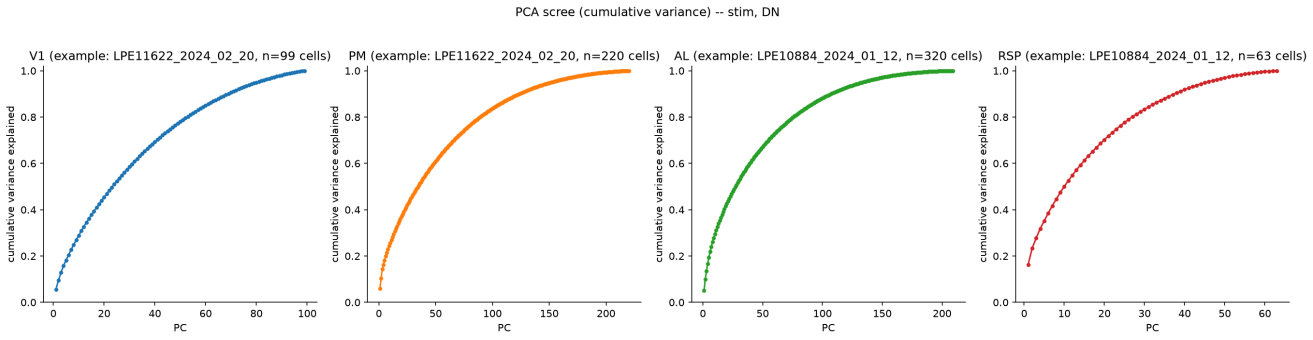
Shuffle comparison stim



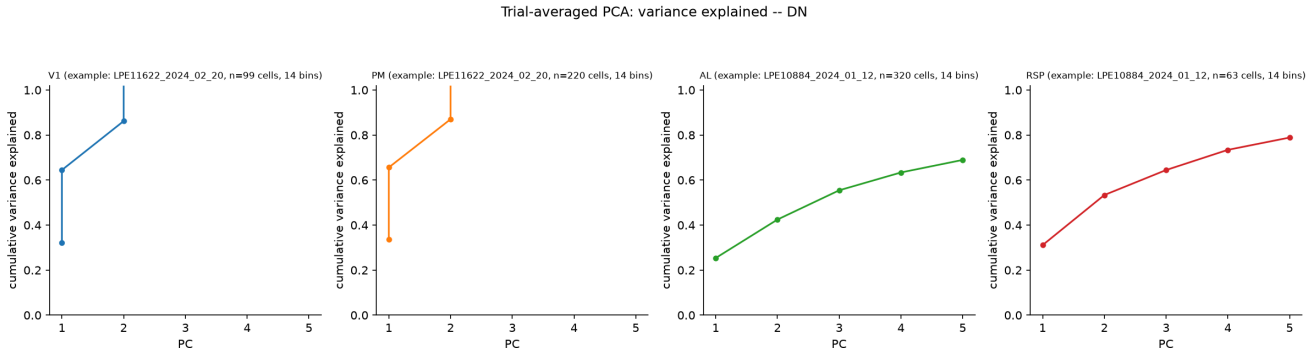
Dimensionality vs size



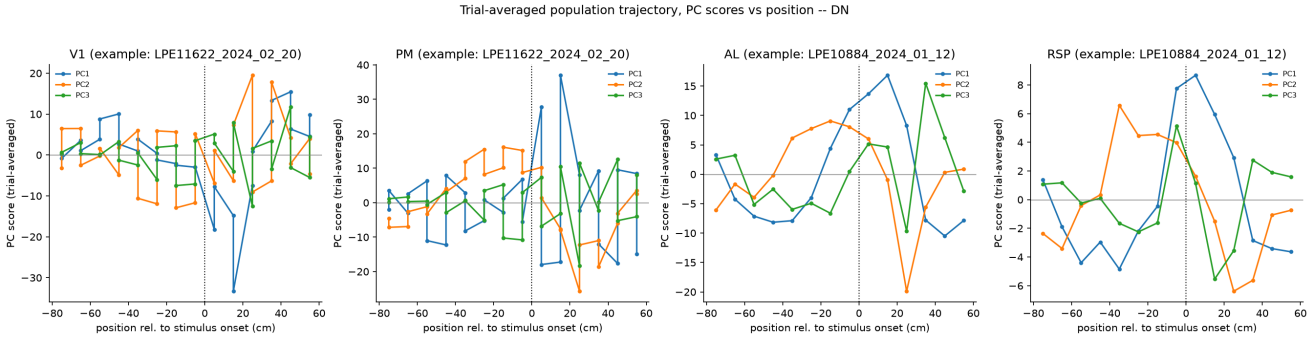
Scree example stim



Trajectory variance explained



Trajectory pc vs position



Trajectory pc1 pc2

